Post hoc bounds for heterogeneous tests

Romain Périer

joint work with Gilles Blanchard, Sebastian Döhler, Guillermo Durand and Étienne Roquain.

Context

Multiple testing setting

- $\mathcal{P}$ a set of probability measures, data $X \sim \mu \in \mathcal{P}$,
- $H_1, \dots, H_m \subset \mathcal{P}$, m null hypotheses,
- For all H_i, we have a p-value $p_i(X)$.

Notations :

- $\mathcal{H}_0(\mu) = \{i : \mu \in H_i\}$,
- $m_0 = |\mathcal{H}_0(\mu)|$,
- $V(S) = |\mathcal{H}_0(\mu) \cap S|$.

Heterogeneous setting

- $\forall \mu \in \mathcal{P}, \forall i \in \mathcal{H}_0(\mu), \forall \alpha \in [0,1], \mathbb{P}(p_i \leq \alpha) \leq F_i(\alpha)$,
- $\forall \mu \in \mathcal{P}$, $(p_i)_{i \in \mathcal{H}_0(\mu)}$ are independent.

Aim : Use the knowledge on the F_i to gain power.

Ex. of heterogeneous settings

Binomial tests

Let $(X_i)_{1 \leq i \leq m}$ be independent variables such that, under H_i, $X_i \sim \mathcal{B}(n_i, \pi_i)$.

- $p_i := \mathbb{P}_{Y \sim \mathcal{B}(n_i, \pi_i)}(Y \leq X_i) = \sum_{j=0}^{X_i} \binom{n_i}{j} \pi_i^j (1 - \pi_i)^{n_i - j}$,
- p_i is a discrete p-value and if the couples (n_i, π_i) are different, the F_i are.

Other examples

- Fisher's exact tests,
- Genome-wide association study.

Post hoc bounds

Post hoc bound

A post hoc bound of level $\alpha \in [0,1]$ is a statistic $\hat{V}_\alpha : \mathcal{P}(\mathbb{N}_m) \to \mathbb{N}$ such that, for all $\mu \in \mathcal{P}$,

$$\mathbb{P}_{X \sim \mu}\Big(\forall S \subset \mathbb{N}_m, V(S) \leq \hat{V}_\alpha(S)\Big) \geq 1 - \alpha.$$

- Allows user to choose freely their rejection set.

Joint Error Rate

For a reference family $\mathcal{R} = \big((R_k, \zeta_k)\big)_{1 \leq k \leq K}$ and $\mu \in \mathcal{P}$, the joint error rate is :

$$JER(\mathcal{R}, \mu) := \mathbb{P}_{X \sim \mu}\big(\exists k \in \mathbb{N}_K : V(R_k) > \zeta_k\big).$$

- Post hoc bounds ideas restricted at reference regions,
- Allows to compute a valid post hoc bound with interpolation (Blanchard et al., 2020, Durand et al., 2020).

Top-k setting :

$$R_k = \big\{\sigma(i), 1 \leq i \leq k\big\} \text{ (where } p_{\sigma(1)} \leq \dots \leq p_{\sigma(m)}\text{)}.$$

References

Blanchard, Gilles, Pierre Neuvial, and Etienne Roquain (2020). "Post hoc confidence bounds on false positives using reference families". In: *Ann. Statist.* 48.3.

Durand, Guillermo, Gilles Blanchard, Pierre Neuvial, and Etienne Roquain (2020). "Post hoc false positive control for structured hypotheses". In: *Scand. J. Stat.* 47.4.

Genovese, Christopher R. and Larry Wasserman (2006). "Exceedance control of the false discovery proportion". In: *J. Amer. Statist. Assoc.* 101.476.

Périer, Romain, Gilles Blanchard, Guillermo Durand, Sebastian Döhler, and Étienne Roquain (2025 - to appear). "Confidence envelopes for the false discoveries with heterogeneous data".

Reeve, Henry W J (2024). *A short proof of the Dvoretzky–Kiefer–Wolfowitz–Massart inequality.* arXiv: `2403.16651 [math.PR]`.

Shorack, Galen R. and Jon A. Wellner (1986). *Empirical processes with applications to statistics.* Wiley Series in Probability and Mathematical Statistics: Probability and Mathematical Statistics. John Wiley & Sons, Inc., New York. ISBN: 0-471-86725-X.

Concentration inequalities

DKWM inequality - Improved by Reeve, 2024

Let $X_1, \dots, X_m \overset{iid}{\sim} J$ and $\hat{J}_m(t) := \frac{1}{m} \sum_{1 \leq i \leq m} \mathbb{1}_{X_i \leq t}$ for all $t \in \mathbb{R}$.

For all $\lambda \geq 0$,

$$\mathbb{P}\Big(\sqrt{m} \sup_{t \in \mathbb{R}} \Big(\hat{J}_m - J\Big)(t) > \lambda\Big) \leq \exp(-2\lambda^2).$$

- Compare $\hat{J}_m$ with a shared cdf.

Bretagnolle's inequality, Shorack and Wellner, 1986 with improvement from Reeve, 2024

Let $X_1, \dots, X_m$ be independent, denote by J_i the cdf of X_i and $\bar{J}$ the mean of J_i's.

Then, for all $\lambda \geq 0$,

$$\mathbb{P}\Big(\sqrt{m} \sup_{t \in \mathbb{R}} \Big(\hat{J}_m - \bar{J}\Big)(t) > \lambda\Big) \leq e \cdot \exp(-2\lambda^2).$$

- Compare $\hat{J}_m$ with $\bar{J}$.

Top-k calibrations

- Let $\alpha \in (0,1]$ and $\lambda_\alpha = \sqrt{\frac{1 + \log(1/\alpha)}{2}}$.

Bretagnolle's top-k calibration

Let

$$\zeta_k = k \wedge \Big\lfloor m\bar{F}(p_{(k)}) + \sqrt{m}\lambda_\alpha \Big\rfloor.$$

Then, the reference family $\mathfrak{R} = \big((R_k, \zeta_k)\big)_{k \in \mathcal{K}}$ has its JER bounded by α.

- $\mathbb{G}(t) = \big(\mathbb{1}_{\{p_i \leq t\}} - F_i(t)\big)_{1 \leq i \leq m}$,
- $\mathbf{F}(t) := \big(F_i(t)\big)_{1 \leq i \leq m}$.

Adaptive version of Bretagnolle's calibration

Let $\hat{m}_0$ such that

$$\left\{\sup_{t \in [0,1]} \sum_{i \in \mathcal{H}_0} \mathbb{G}_i(t) \leq \sqrt{m_0}\lambda_\alpha\right\} \subset \{m_0 \leq \hat{m}_0\}.$$

Define :

$$\zeta_k = k \wedge \left\lfloor \sum_{1 \leq i \leq \hat{m}_0} \Big(\mathbf{F}(p_{(k)})\Big)_{[i]} + \sqrt{\hat{m}_0}\lambda_\alpha \right\rfloor.$$

Then, the reference family $\mathfrak{R} = \big((R_k, \zeta_k)\big)_{k \in \mathcal{K}}$ has its JER bounded by α.

Aim : Find the best $\hat{m}_0$.

Local test paradigm

- $\forall A \subset \{1, \dots, m\}$, $H_A := \bigcap_{i \in A} H_i$,
- $\forall A \subset \{1, \dots, m\}$, local test φ_A,
- $\mathcal{N}(X) = \big\{A \subset \mathbb{N}_m : \varphi_A(X) = 0\big\}$.

Inversion procedure, Genovese and Wasserman, 2006

Assume that $X \sim \mu$ and $\varphi_{\mathcal{H}_0(\mu)}$ is a test of level α. Then $\hat{V}_\varphi^{\mathrm{IP}}(S) := \max_{A \in \mathcal{N}} |A \cap S|$ is a post hoc bound of level α.

- **Advantage :** Best bound on $V(S)$ from local tests,
- **Drawback :** Exponential computational cost.

Concentration and shortcuts

Suppose : $\forall A \subset \{1, \dots, m\}$, $\varphi_A = \sup_{t \in [0,1]} \varphi_{A,t}$.

Heterogeneous-designed shortcut

$$\hat{V}_\varphi^{\mathrm{IP}}(S) \leq \min_{t \in [0,1]} \left\{ \max \left\{ \ell \in \{0, \dots, |S|\} : \min_{\substack{A \subset [\![1,m]\!] \\ |A \cap S| = \ell}} \varphi_{A,t} = 0 \right\} \right\}.$$

Appli : Bretagnolle's inequality

- $\varphi_{A,t} = \mathbb{1}\left\{\sum_{i \in A} \mathbb{G}_i(t) > \sqrt{|A|}\lambda\right\}$,
- $\min_{\substack{A \subset [\![1,m]\!] \\ |A \cap S| = \ell}} \varphi_{A,t} = \min_{0 \leq j \leq m - |S|} \mathbb{1}\left\{\sum_{1 \leq i \leq \ell} \big(\mathbb{G}(t)\big)_{(i:S)} + \sum_{1 \leq i \leq j} \big(\mathbb{G}(t)\big)_{(i:S^c)} > \lambda\sqrt{\ell + j}\right\}$.

Bretagnolle's upper bound on m_0

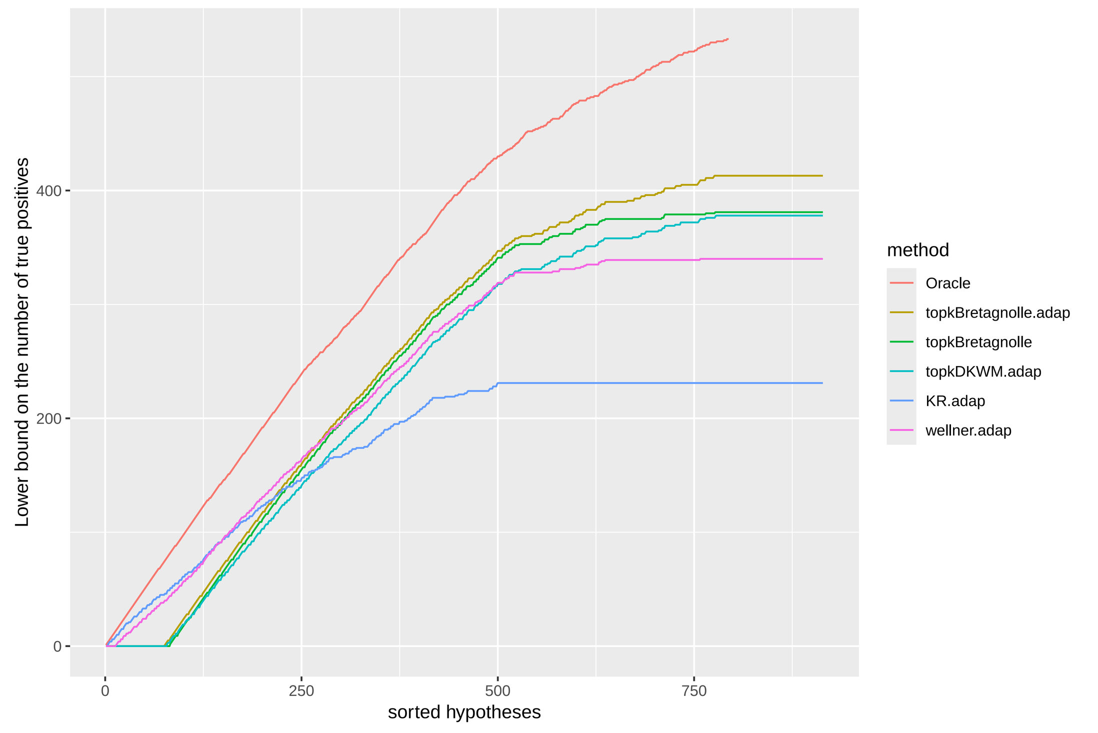
Define, for all $t \in [0,1]$,

$$b(t) = \max\left\{\ell : \ell^{-1/2}\left(\sum_{1 \leq j \leq \ell} \big(\mathbb{G}(t)\big)_{(j)}\right) \leq \lambda_\alpha\right\}.$$

Then, within the event $\left\{\sup_{t \in [0,1]} \sum_{i \in \mathcal{H}_0} \mathbb{G}_i(t) \leq \sqrt{m_0}\lambda_\alpha\right\}$, we have $m_0 \leq \hat{V}_\varphi^{\mathrm{IP}}\big([\![1,m]\!]\big) \leq \min_{t \in [0,1]} b(t) = \min_{i \in [\![0,m]\!]} b(p_i)$.

Simulations

Random support setting



Fisher's exact test setting